CYBERSECURITY INTERNSHIP

WEEK 1 | Learning, Labs & Writeups

DG Interns Hub

Prepared by: CHINTALAPUDI VENKATA SIVA SAI RAJDEEP

BUILD • PRACTISE • DOCUMENT

Week 1 at a glance

The goal was to build a practical foundation in cybersecurity, Linux, and networking.

Cybersecurity basics

CIA Triad, security roles, common threats, and ethical practice.

Hands-on practice

Kali Linux, Ubuntu, VirtualBox, host-only networking, and Linux commands.

Learning platforms

OverTheWire Bandit, TryHackMe topics, notes, and writeups.

Focus: learn the concept → practise it → record evidence.

Cybersecurity fundamentals

Core ideas that guide how systems and information are protected.

CIA Triad

Confidentiality — only authorised access

Integrity — data stays accurate

Availability — services are accessible

Threats & risks

Threat: something that can cause harm

Vulnerability: a weakness

Risk: likelihood and impact of harm

Security roles

SOC Analyst: monitors and investigates

Pen Tester: tests with permission

Incident Responder: helps contain incidents

Ethical rule: only test systems when you have clear permission and scope.

Oracle VirtualBox Manager

File Machine Help

New Open Settings Discard Start

kali-linux-2026.2-virtualbox-amd64
Powered Off

ubuntu 26.04
Powered Off

Details

General

Name: kali-linux-2026.2-virtualbox-amd64
Operating System: Debian (64-bit)

System

Base Memory: 4986 MB
Processors: 2
Boot Order: Hard Disk, Optical
Acceleration: Nested Paging, PAE/NX, KVM Paravirtualization

Preview



Display

Video Memory: 128 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled

Storage

Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: kali-linux-2026.2-virtualbox-amd64.vdi (Normal, 80.09 GB)

Audio

Host Driver: Windows DirectSound
Controller: ICH AC97

Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT)

USB

USB Controller: OHCI
Device Filters: 0 (0 active)

Shared folders

None

Description

Kali Rolling (2026.2) x64
2026-06-16

28°C Search ENG 100% 18:35

Linux practical work

Practised commands and basic system administration.

Commands

Practised 30 Linux commands for navigation, files, and system information.

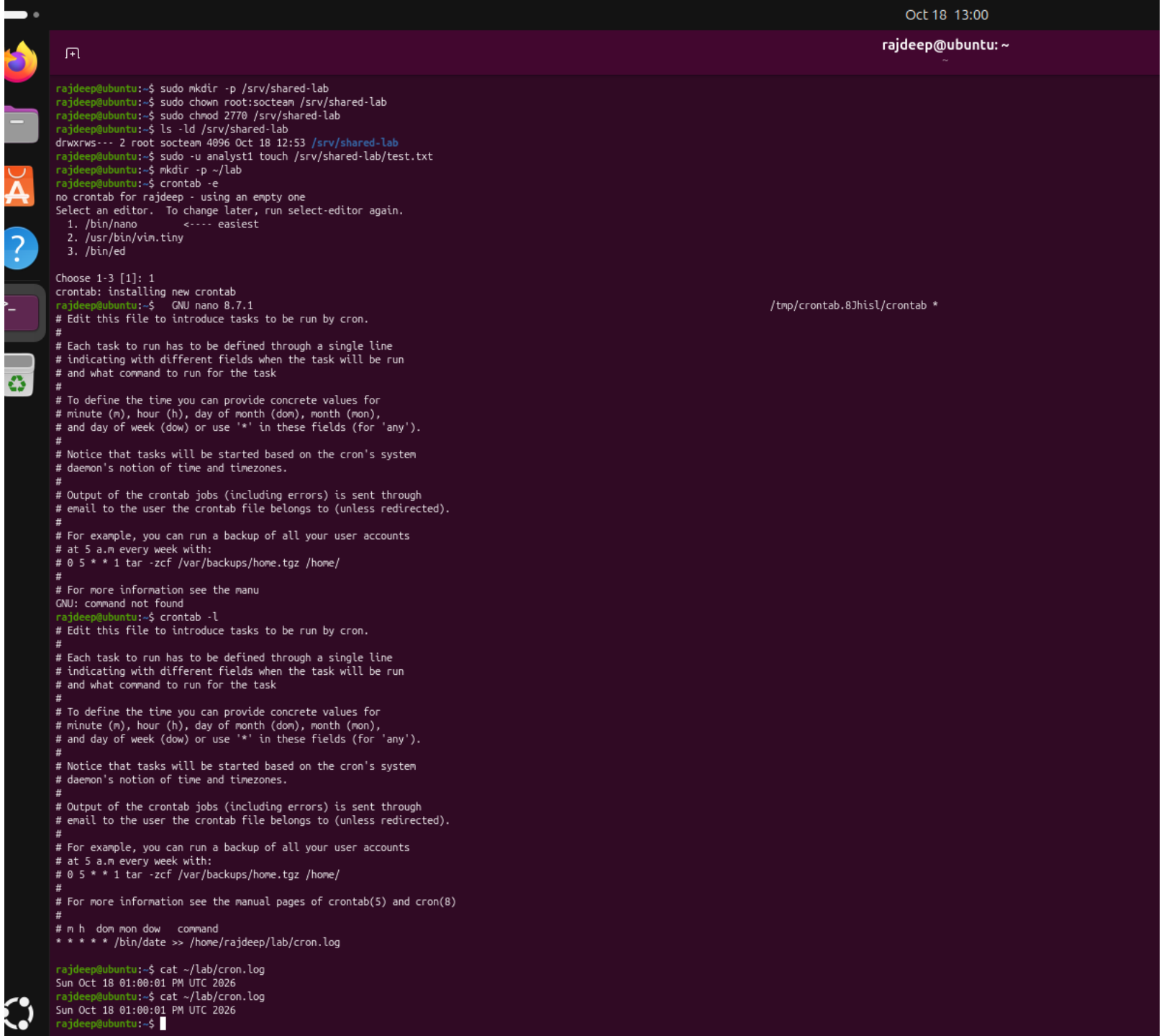
Users & permissions

Created users and groups, then tested access to a shared directory.

Scheduled task

Configured a cron job to append date/time output to a log file.

Key learning: use least privilege and verify permissions rather than assuming access is correct.


Oct 18 13:00
rajdeep@ubuntu: ~
rajdeep@ubuntu:~$ sudo mkdir -p /srv/shared-lab
rajdeep@ubuntu:~$ sudo chown root:socteam /srv/shared-lab
rajdeep@ubuntu:~$ sudo chmod 2770 /srv/shared-lab
rajdeep@ubuntu:~$ ls -ld /srv/shared-lab
drwxrws--- 2 root socteam 4096 Oct 18 12:53 /srv/shared-lab
rajdeep@ubuntu:~$ sudo -u analyst1 touch /srv/shared-lab/test.txt
rajdeep@ubuntu:~$ mkdir -p ~/lab
rajdeep@ubuntu:~$ crontab -e
no crontab for rajdeep - using an empty one
Select an editor. To change later, run select-editor again.
1. /bin/nano <---- easiest
2. /usr/bin/vim.tiny
3. /bin/ed
Choose 1-3 [1]: 1
crontab: installing new crontab
rajdeep@ubuntu:~$ GNU nano 8.7.1 /tmp/crontab.8Jhisl/crontab *
Edit this file to introduce tasks to be run by cron.
#
Each task to run has to be defined through a single line
indicating with different fields when the task will be run
and what command to run for the task
#
To define the time you can provide concrete values for
minute (m), hour (h), day of month (dom), month (mon),
and day of week (dow) or use '*' in these fields (for 'any').
#
Notice that tasks will be started based on the cron's system
daemon's notion of time and timezones.
#
Output of the crontab jobs (including errors) is sent through
email to the user the crontab file belongs to (unless redirected).
#
For example, you can run a backup of all your user accounts
at 5 a.m every week with:
0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
For more information see the manu
GNU: command not found
rajdeep@ubuntu:~$ crontab -l
Edit this file to introduce tasks to be run by cron.
#
Each task to run has to be defined through a single line
indicating with different fields when the task will be run
and what command to run for the task
#
To define the time you can provide concrete values for
minute (m), hour (h), day of month (dom), month (mon),
and day of week (dow) or use '*' in these fields (for 'any').
#
Notice that tasks will be started based on the cron's system
daemon's notion of time and timezones.
#
Output of the crontab jobs (including errors) is sent through
email to the user the crontab file belongs to (unless redirected).
#
For example, you can run a backup of all your user accounts
at 5 a.m every week with:
0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
For more information see the manual pages of crontab(5) and cron(8)
#
m h dom mon dow command
* * * * * /bin/date >> /home/rajdeep/lab/cron.log
rajdeep@ubuntu:~$ cat ~/lab/cron.log
Sun Oct 18 01:00:01 PM UTC 2026
rajdeep@ubuntu:~$ cat ~/lab/cron.log
Sun Oct 18 01:00:01 PM UTC 2026
rajdeep@ubuntu:~$

Hands-on learning platforms

Command-line practice and beginner cybersecurity learning.

OverTheWire Bandit

Worked through levels up to Level 10.

Practised reading files, using shell commands, and solving command-line challenges.

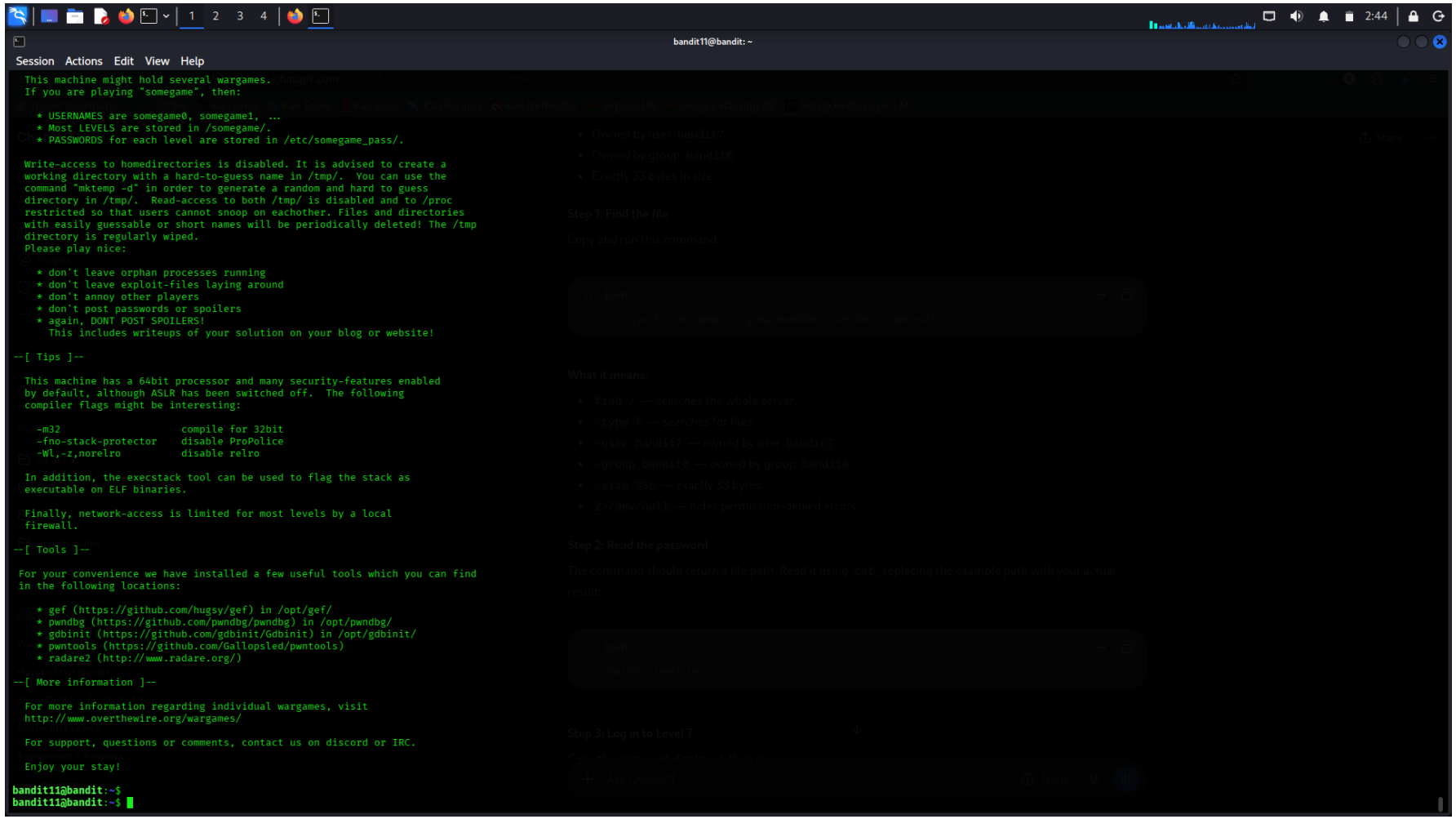
Prepared a combined writeup.

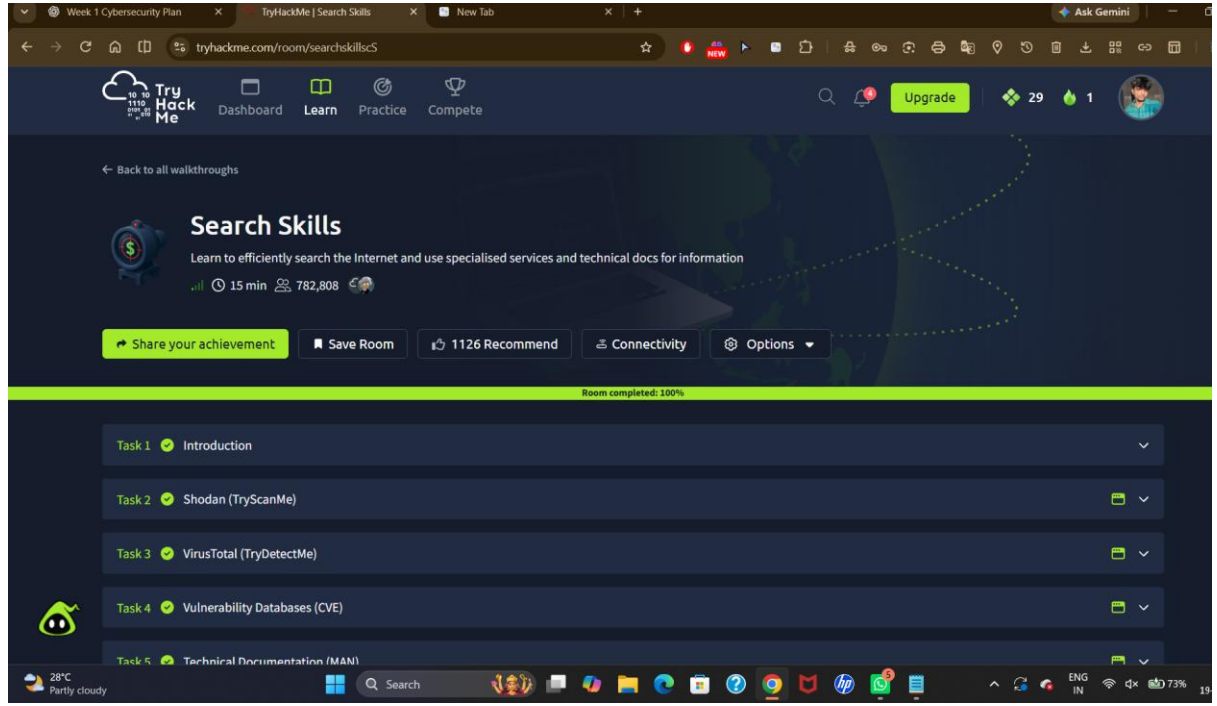
TryHackMe

Prepared writeups for beginner topics including search skills, Linux, networking, LANs, and the Pyramid of Pain.

Add exact task answers and screenshots from rooms you completed.

Do not include passwords or other sensitive information in reports or screenshots.


1 2 3 4
2:44
bandit11@bandit: ~
Session Actions Edit View Help
This machine might hold several wargames.
If you are playing "somegame", then:
* USERNAMES are somegame0, somegame1, ...
* Most LEVELS are stored in /somegame/.
* PASSWORDS for each level are stored in /etc/somegame_pass/.
Write-access to homedirectories is disabled. It is advised to create a
working directory with a hard-to-guess name in /tmp/. You can use the
command "mktemp -d" in order to generate a random and hard to guess
directory in /tmp/. Read-access to both /tmp/ is disabled and to /proc
restricted so that users cannot snoop on eachother. Files and directories
with easily guessable or short names will be periodically deleted! The /tmp
directory is regularly wiped.
Please play nice:
* don't leave orphan processes running
* don't leave exploit-files laying around
* don't annoy other players
* don't post passwords or spoilers
* again, DONT POST SPOILERS!
This includes writeups of your solution on your blog or website!
--[Tips]--
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:
-m32 compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro disable relro
In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.
Finally, network-access is limited for most levels by a local
firewall.
--[Tools]--
For your convenience we have installed a few useful tools which you can find
in the following locations:
* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)
--[More information]--
For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/
For support, questions or comments, contact us on discord or IRC.
Enjoy your stay!
bandit11@bandit:~$
bandit11@bandit:~$


Week 1 Cybersecurity Plan
TryHackMe | Search Skills
New Tab
Ask Gemini
tryhackme.com/room/searchskillscS
Try Hack Me
Dashboard
Learn
Practice
Compete
Upgrade
29
1
Back to all walkthroughs
Search Skills
Learn to efficiently search the Internet and use specialised services and technical docs for information
15 min
782,808
Share your achievement
Save Room
1126 Recommend
Connectivity
Options
Room completed: 100%
Task 1 Introduction
Task 2 Shodan (TryScanMe)
Task 3 VirusTotal (TryDetectMe)
Task 4 Vulnerability Databases (CVE)
Task 5 Technical Documentation (MAN)
28°C
Partly cloudy
Search
ENG
IN
73%

Key learnings & challenges

What improved during the first week.

Key learnings

- Linux terminal and file permissions
- IP addresses, ports, protocols, and LANs
- Virtual machine networking
- Documenting steps and evidence

Challenges & response

- Faced setup and command-line issues
- Checked configuration and read error messages
- Retested commands and connectivity
- Learned to troubleshoot step by step

Add a real screenshot of your own lab work to strengthen this slide.

WEEK 1 SUMMARY

A strong start in cybersecurity fundamentals.

- Built a basic Linux and networking lab.
- Practised command-line and account-permission tasks.
- Worked through Bandit and beginner learning materials.
- Documented learning through notes, writeups, and evidence.

Next: continue Linux and networking practice, then build SOC-focused monitoring skills.